

Phylogenetic Analysis Using NCBI and Galaxy

GROUP 4 - Anatidae
BIO 300 - Cell & Molecular Biology

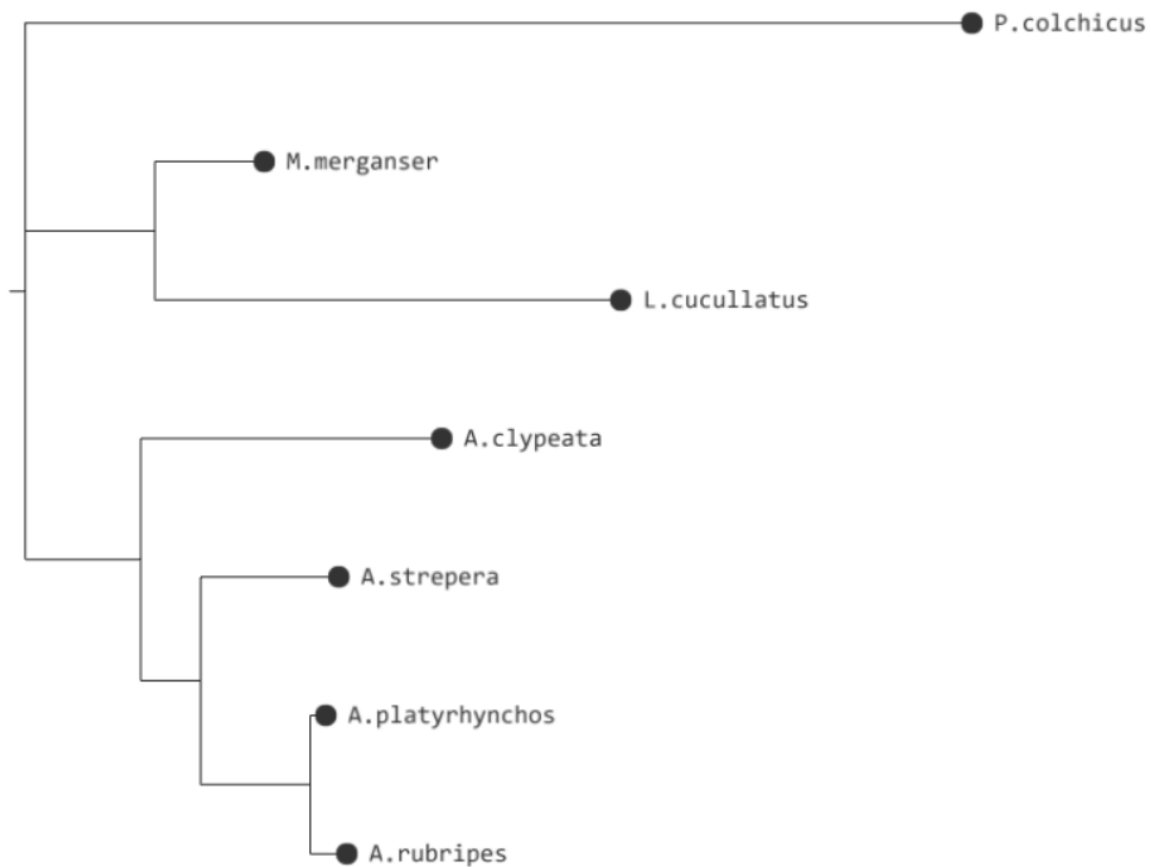
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PhyloCanvas.js

Figure 1. Phylogenetic tree of the selected Anatidae species based on mitochondrial COI sequences generated using FastTree.

```
(P.colchicus:0.206692325,(M.merganser:0.023839098,L.cucullatus:0.101697343)0.982:0.028336913,
(A.clypeata:0.065736263,(A.strepera:0.030137754,
(A.platyrhynchos:0.003364787,A.rubripes:0.007982354)0.996:0.023935695)0.855:0.013115456)0.945:0.025194752)
;
```

Figure 2. Tree file generated by FASTTREE from the aligned mitochondrial COI sequences of the selected Anatidae species and the outgroup, *Phasianus colchicus*.

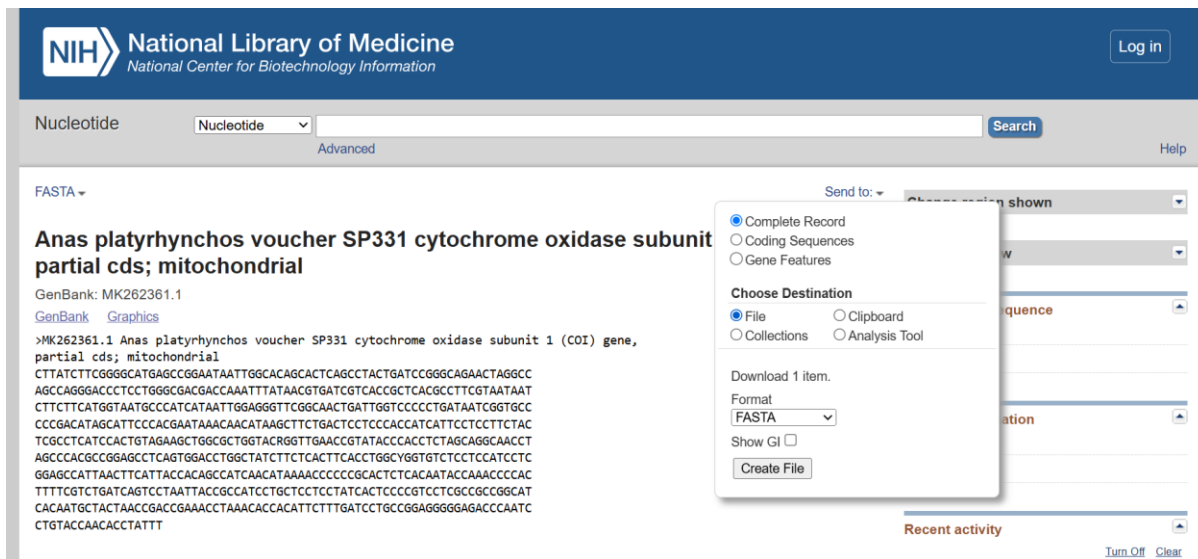


Figure 3. Workflow of how sequence data was collected from NCBI Nucleotide. Searching for each species using its scientific name and the COI/COX1 gene. Checking each sequence to make sure it covers the correct region and length. Saving the original, unchanged FASTA files with clear names for analysis.

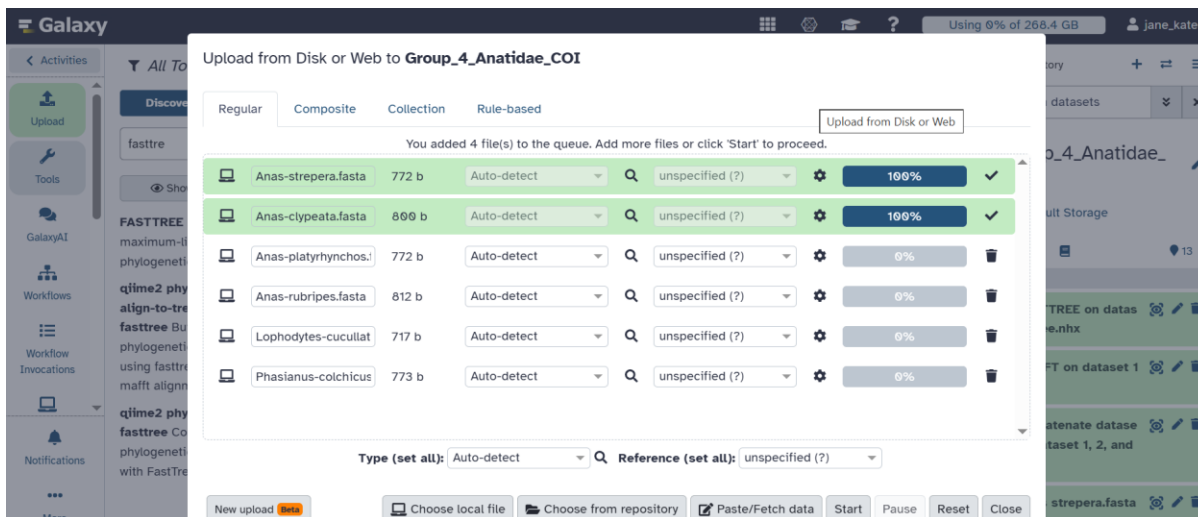


Figure 4. Uploading and preparing FASTA files in Galaxy. Downloaded COI sequences from NCBI were uploaded into the Galaxy history. Each file was checked to confirm the correct FASTA format and renamed with clear species labels for analysis.

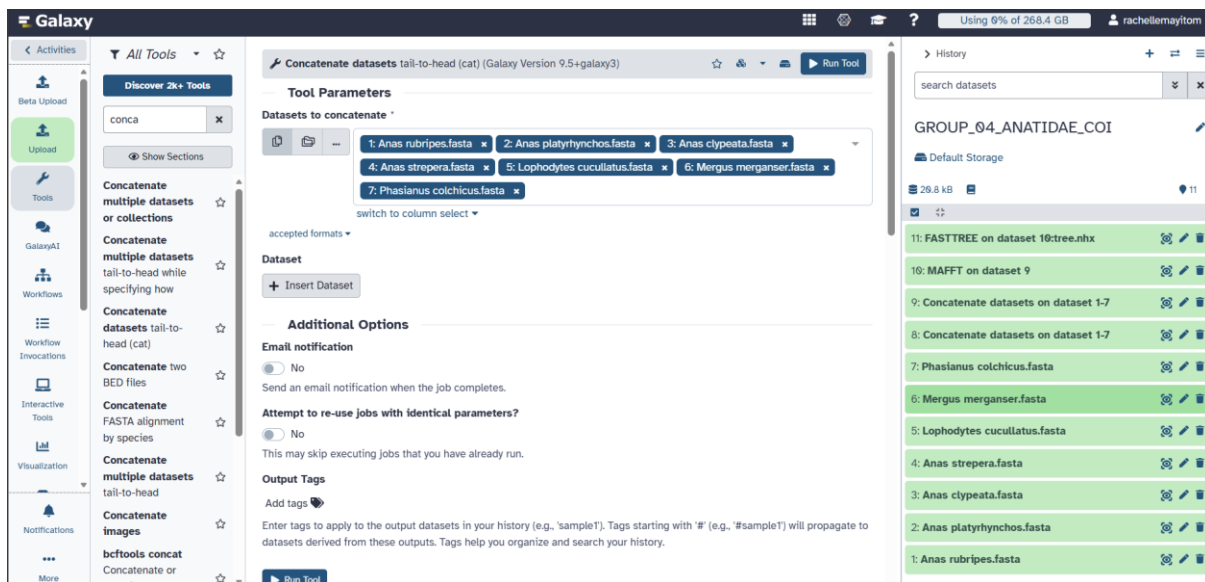


Figure 5. Combining sequences into a multi-FASTA file in Galaxy. Concatenating individual COI FASTA records into a single dataset.

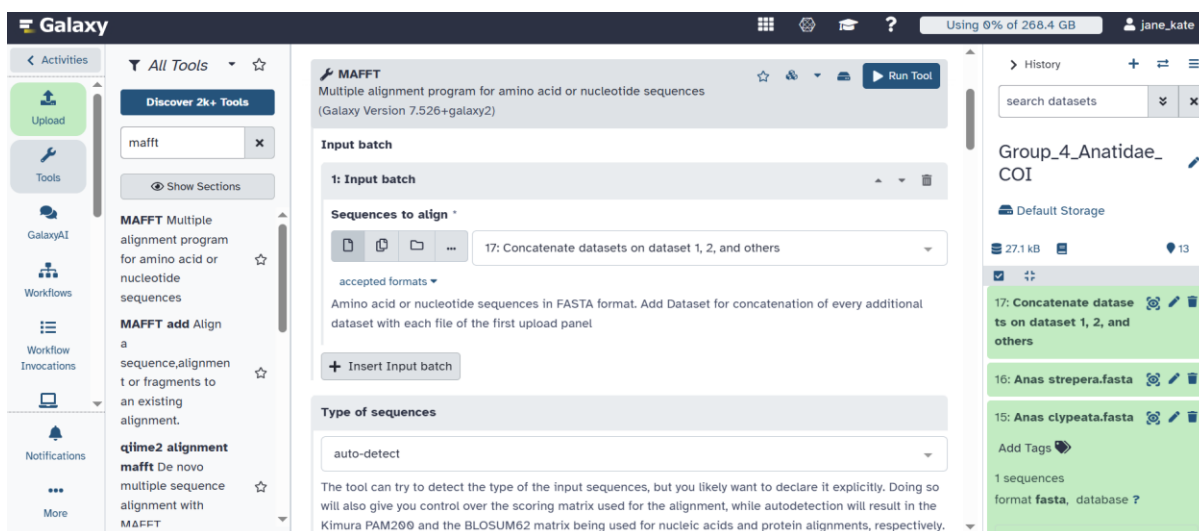


Figure 6. Running the MAFFT multiple sequence alignment tool in Galaxy. The selected COI sequences were aligned using MAFFT to identify conserved and variable nucleotide positions among the species.

	Id	Actual Id	Begin	End	Link	Real Length	Aligned Leng...	Left
1	P.colchicus	P.colchicus	1	648		648	648	5
2	M.merganser	M.merganser	1	648		648	648	5
3	L.cucullatus	L.cucullatus	1	590		590	590	102
4	A.strepera	A.strepera	1	648		648	648	5
5	A.clypeata	A.clypeata	1	675		675	675	17
6	A.platyrhynchos	A.platyrhynchos	1	648		648	648	5
7	A.rubripes	A.rubripes	1	692		692	692	0

Figure 7. Visualization of the MAFFT multiple sequence alignment. Alignment results and sequence logos were generated in Galaxy to inspect conserved regions, variations, and sequence similarities across all included species.

The screenshot displays the Galaxy web interface for the FASTTREE tool. The tool parameters are configured as follows:

- Aligned sequences file (FASTA or Phytip format):** fasta
- FASTA file:** 18: MAFFT on dataset 17
- accepted formats:** dropdown menu
- Allow spaces and other restricted characters (but not ') in sequence and quote names in the output tree:** No
- Set starting tree(s):** No starting trees

The right sidebar shows a history of datasets, including 'Group_4_Anatidae_COI' and '18: MAFFT on dataset 17'.

Figure 8. Construction of the phylogenetic tree using FastTree in Galaxy. The aligned COI sequences were analyzed using the maximum-likelihood method to infer the evolutionary relationships among the selected species.

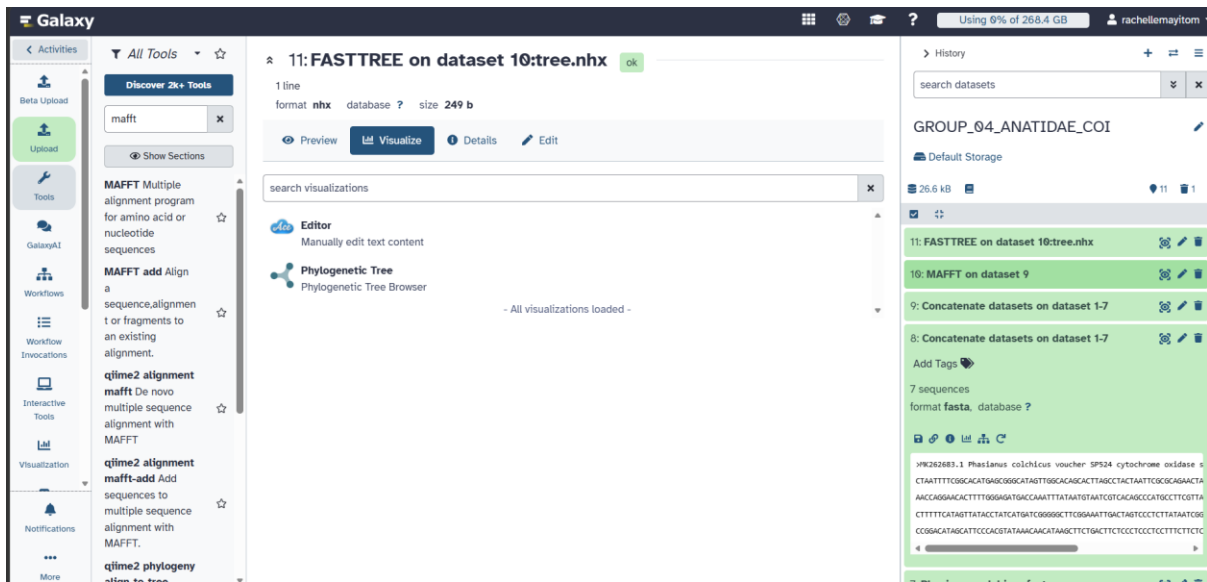


Figure 9. Visualization of the phylogenetic tree generated by *FastTree* in Galaxy, showing the branching pattern and evolutionary relationships among the aligned COI sequences.